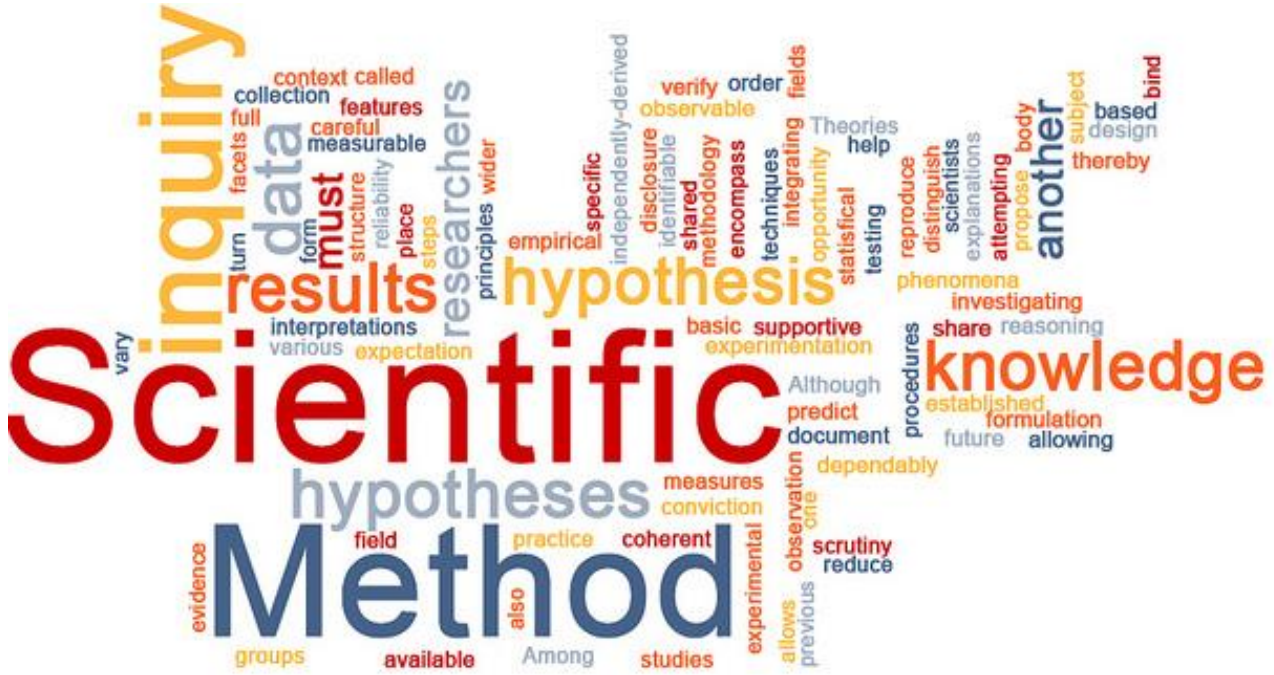



# Thinking like a Psychological Scientist to Understand the Mind

[Dr Yashpal Jogdand]

[AHUL261]

# Overview

- **Introduction**
- Scientific Reasoning and Theories
- Interpreting Research (Correctly)
- Subjectivity and Objectivity in Science
- Conclusion




- **Would people offer to help in this situation?**

# Kitty Genovese case

- Bibb Latané and John Darley (1970) are two social psychologists who taught at universities in New York at the time of Kitty Genovese's prolonged murder, when 38 witnesses failed to call police.

- Paradoxically, they thought, it might be that the greater the number of bystanders who observe an emergency, the less likely any one of them is to help.

- As Latané (1987) put it, “We came up with the insight that perhaps what made the Genovese case so fascinating was itself what made it happen—namely, that not just one or two, but thirty-eight people had watched and done nothing” (p.  78).

# The Number of Bystanders: The Bystander Effect

Latané and Darley (1970) found that in terms of receiving help, there is no safety in numbers.

Dozens of other studies, conducted in the laboratory and in the field, have found what they found: The greater the number of bystanders who witness an emergency, the less likely any one of them is to help the victim.

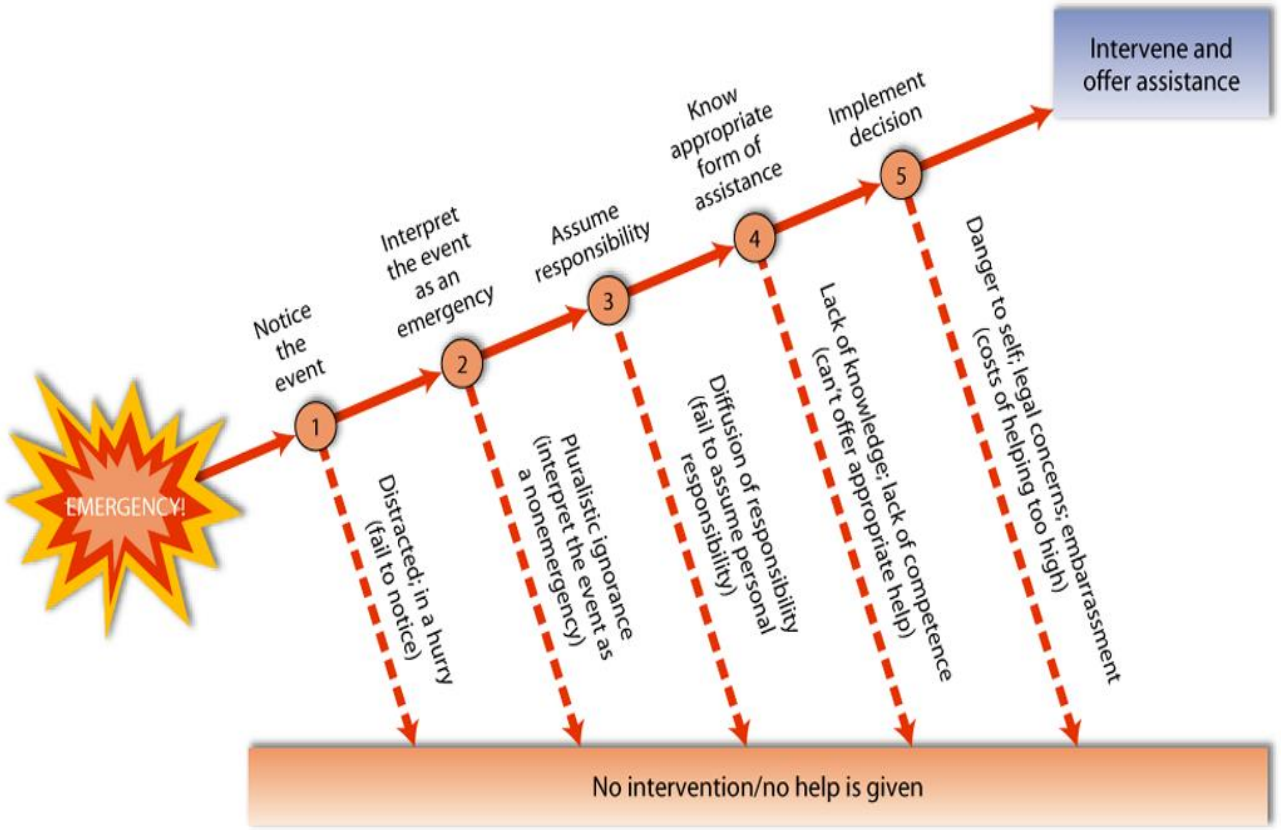
This is known as the **bystander effect.**



https://www.youtube.com/watch?v=BdpdUbW8vbw

## Why is it that people are less likely to help when others are present?

Latané and Darley (1970) developed a step-by-step description of how people decide whether to intervene in an emergency. Part of this description is an explanation of how the number of bystanders can make a difference.



# Interpreting the Event as an Emergency

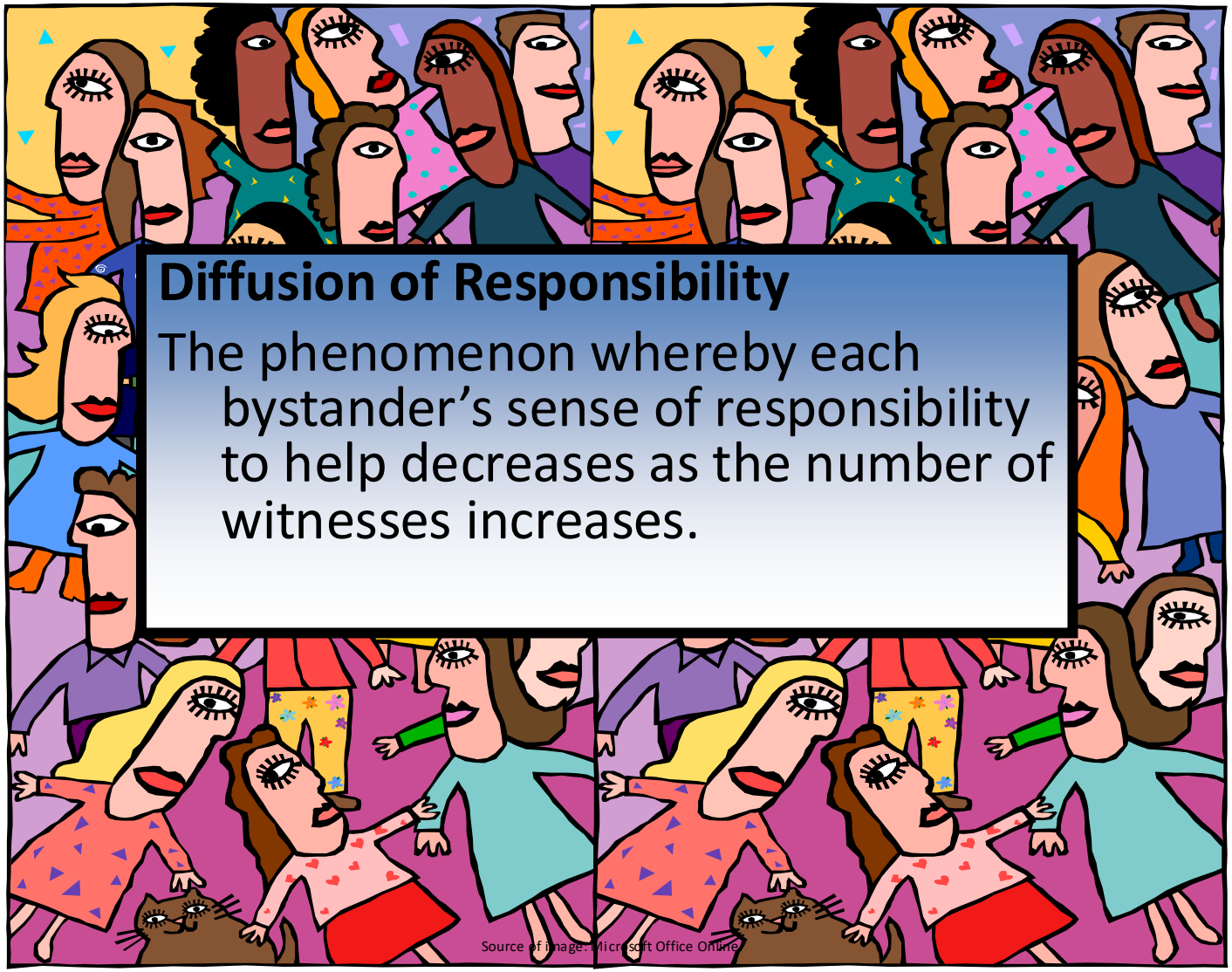


- A key determinant of helping is whether the bystander interprets the event as an emergency—as a situation where help is needed.
- If people assume that nothing is wrong when an emergency is taking place, they will not help.
- When other bystanders are present, people are more likely to assume that an emergency is something innocuous.

**Pluralistic Ignorance**

Bystanders' assuming that nothing is wrong in an emergency because no one else looks concerned.

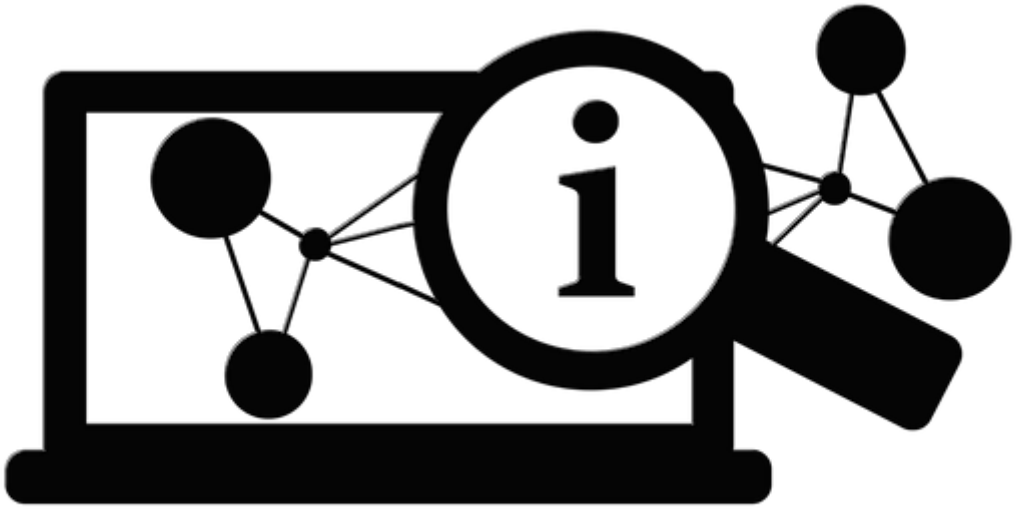
# Assuming Responsibility



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# Scientific Reasoning & Theories



Testing **hypotheses** based on **samples** using **inductive reasoning**

# Inductive Reasoning

What comes next?



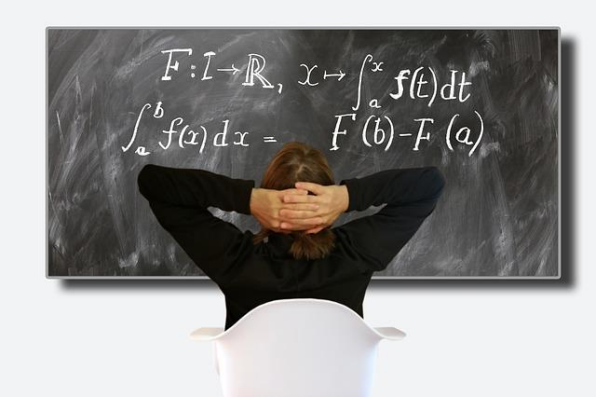
But what if you knew this?





# Features of Good Scientific Theories

- **Accuracy:** Explanations and theories match real-world observations
- E.g. Although people say, “opposites attract,” theories that focus on the role of partner similarity do a better job of explaining the observed data






# Features of Good Scientific Theories

- **Consistency:** A theory has few exceptions and shows agreement with other theories within and across disciplines.

- E.g. The theory of evolution explains many findings across biology and psychology predicting, for example, that humans are better able to solve problems presented in concrete rather than abstract terms

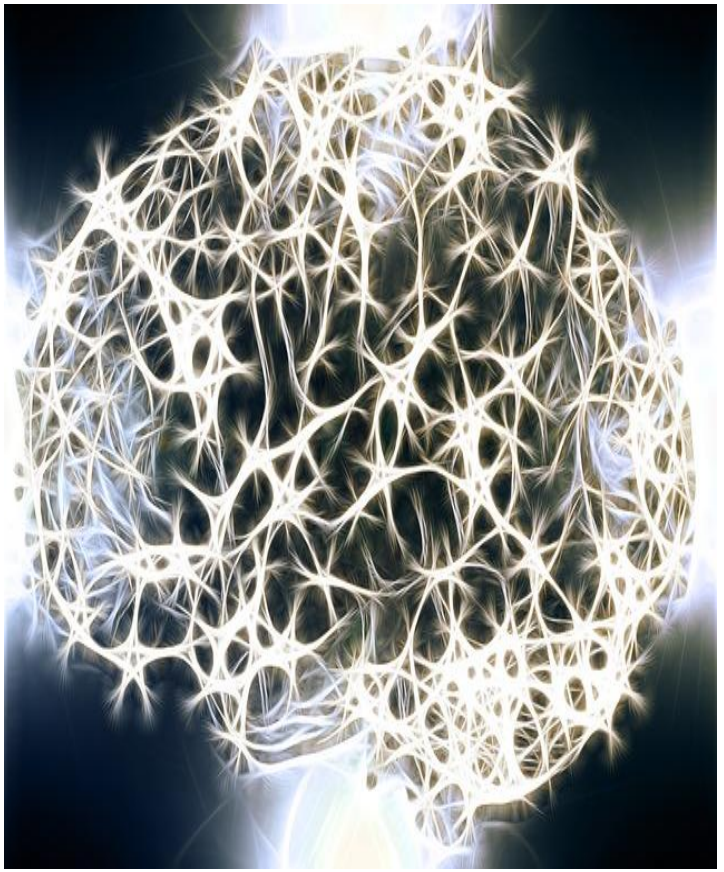


# Features of Good Scientific Theories

**Scope:**

Extent to which a theory extends beyond currently available data, explaining a wide array of phenomena.

E.g. There is a theory that people use mental “short cuts” when making decision rather than weighing every single piece of evidence. This can be seen in consumer purchasing behavior, in romantic relationships, in charitable donations, and in health choices.



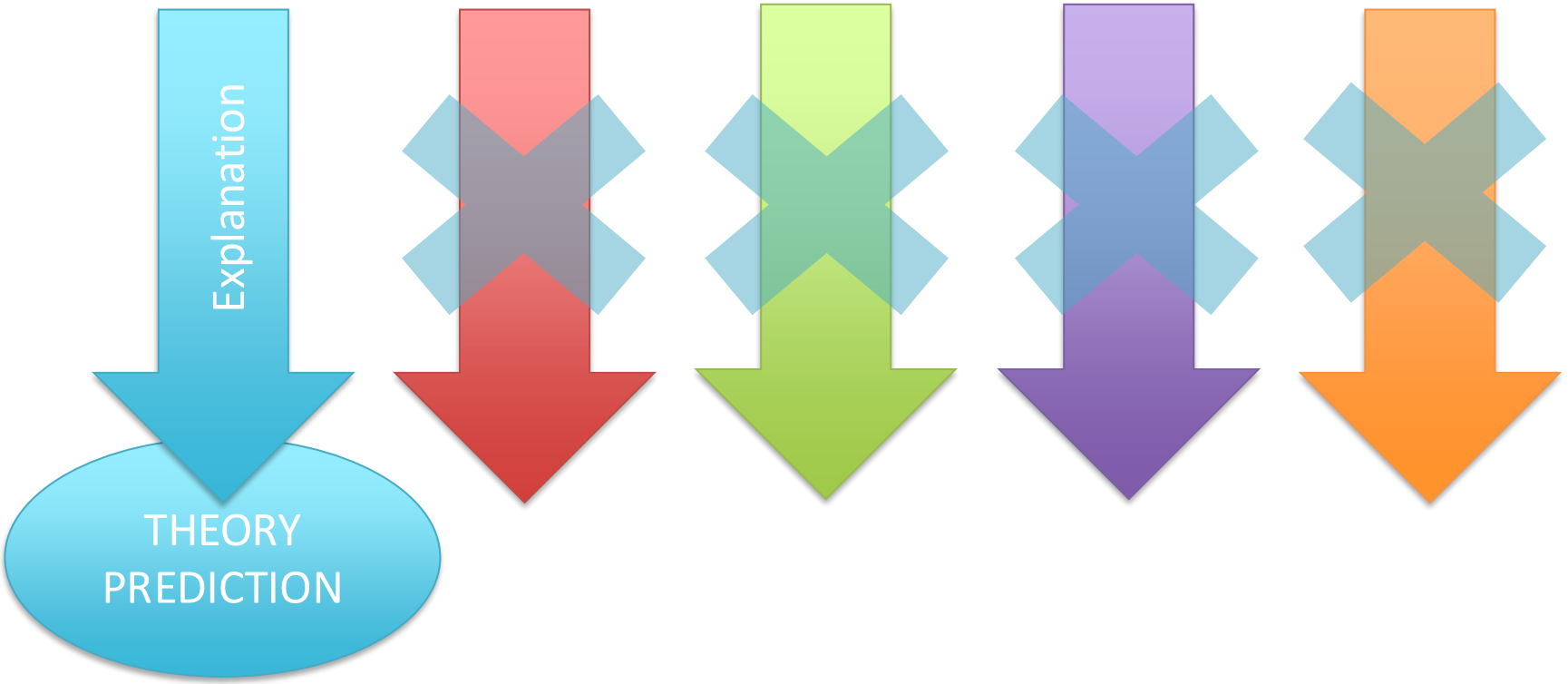
# Features of Good Scientific Theories

- **Simplicity:** When multiple explanations are equally good at explaining the data, the simplest should be selected.
- E.g. The simplest explanation for why "good" people sometimes do "bad" things is because they succumb to some outside influence



# Features of Good Scientific Theories

## Falsifiable



# Which is falsifiable?



- Classical music increases intelligence.
- Having Tea before exams increases performance.
- Mangoes are delicious.
- God does not exist.

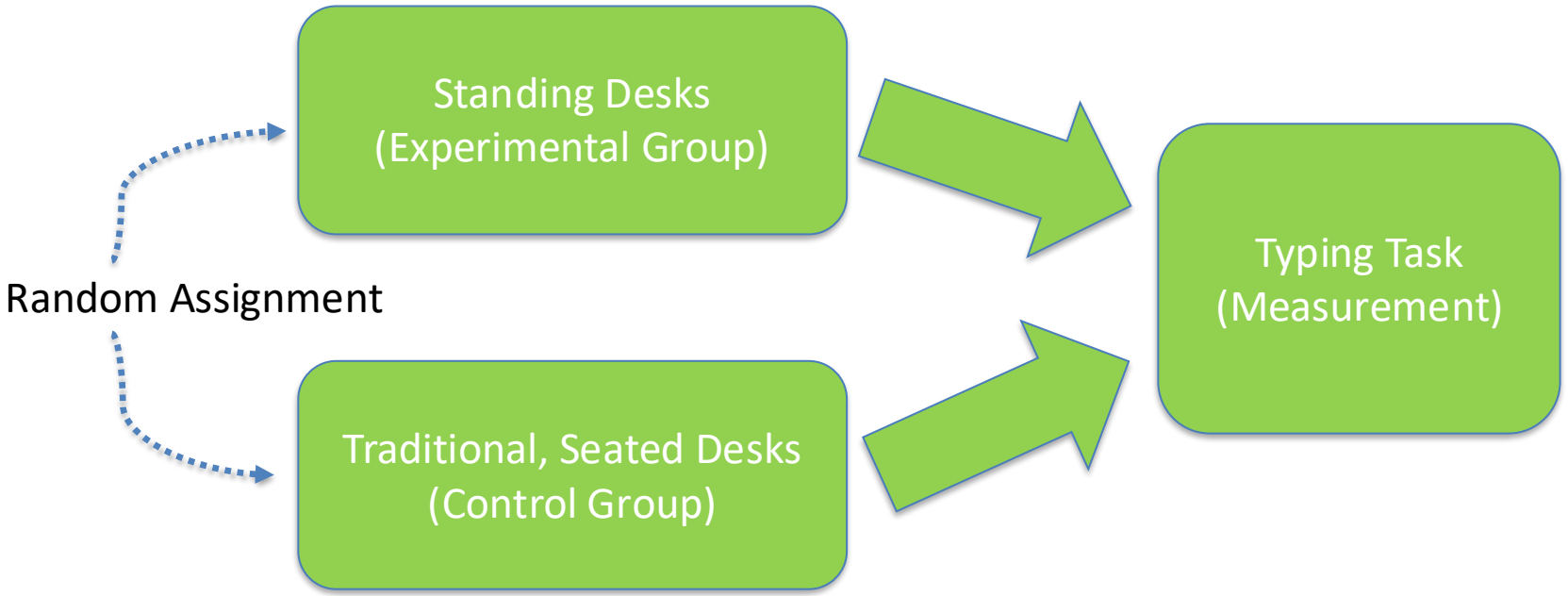
# Overview

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- **Interpreting Research (Correctly)**
- Trusting Science without Proof
- Objectivity in Science
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# Interpreting Research (Correctly)

## Research Hypothesis:

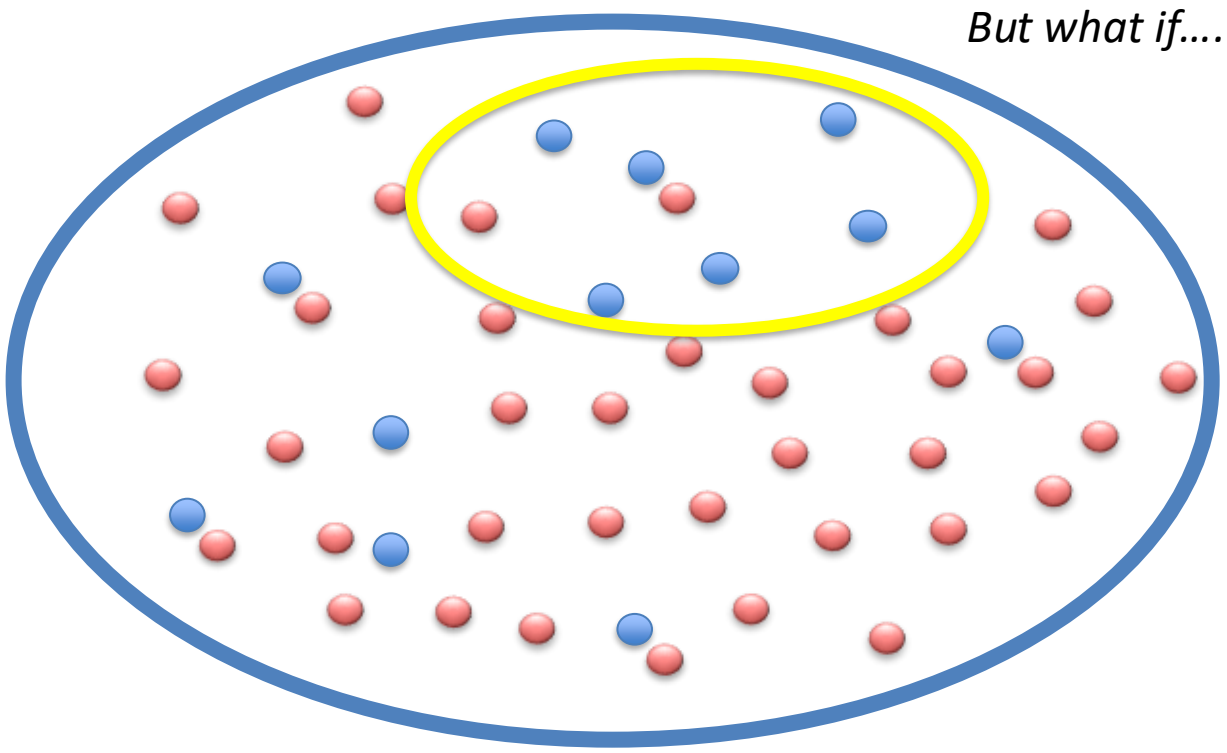
Standing desks increase productivity



# Do standing desks increase productivity?

## 3 Options:

1. Results Support Hypothesis



# Do standing desks increase productivity?

**Options:**

1. Results Support Hypothesis
2. Results Contradict Hypothesis
   a. Sitting > Standing
   b. No difference between groups



# Overview

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# Subjectivity and Objectivity in Science



The image of a scientist as an objective, unbiased observer to the natural world is not accurate. There are too many choices made by the scientist themselves throughout the scientific process that will impact research results.



**Half Full?**

**Half Empty?**

# Discussion

- “fact-value distinction.” Facts are statements about the world and values are our beliefs about the way the world ought to be.

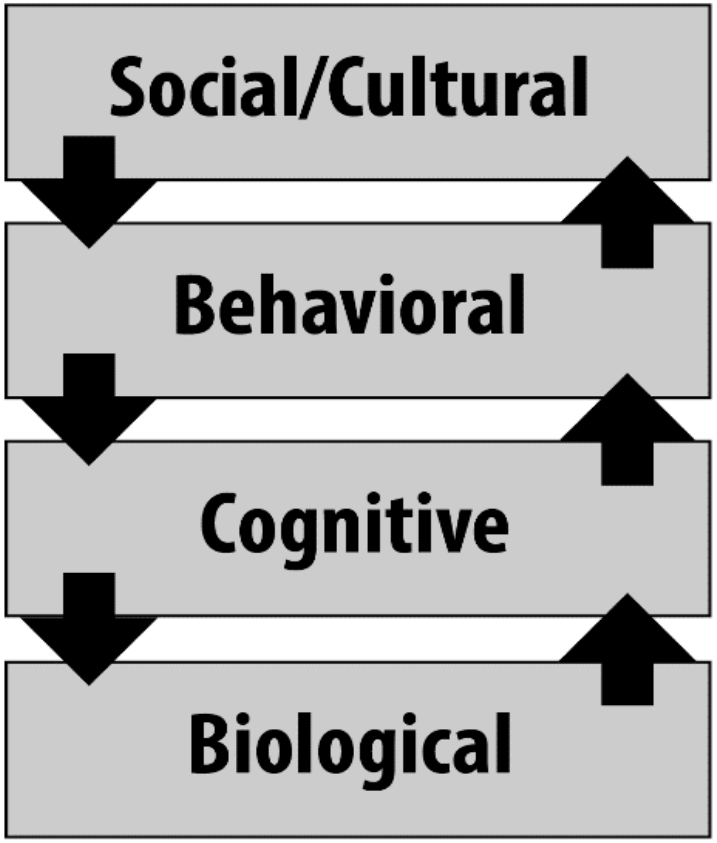
- Our values will inform the kinds of questions that we ask and how we ask them, which will in turn influence our results for any single study.

- Rather than being cold and dispassionate observers, scientists are active in the process of gathering information about the world and interpreting that information to make sense of it (especially in the social sciences).

- As a result, science works best when people with diverse values and backgrounds work collectively to understand complex natural phenomena.

# Diverse Perspectives in Science

Levels of Analysis



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# Q & A

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